

CHARCHIT GANGWAR

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EDUCATION

Vellore Institute of Technology, Bhopal
B.Tech in Computer Science and Engineering
Jingle Bells Public School, Bareilly
12th Grade

CGPA: 9.19
October 2022-Present
Percentage: 92%
July 2021

SKILLS

Programming Languages: C++, Python, SQL

Technologies/Frameworks: HTML, CSS, React.js, Node.js, Express.js, FAST API, Flask, Git, Github, MongoDB, MySQL, RESTful APIs, AWS

Technical Skills: Operating Systems, Database Management, Object-Oriented Programming, Data Structures and Algorithms, Machine Learning

EXPERIENCE

WORKIK

Software Development Engineer Intern

November 2024-March 2025

Skills: Python, Flask, SQLAlchemy, Automation, Webhooks, REST APIs, OAuth, Git

- Architected a centralized backend system connecting 6 developer ecosystems, including GitHub, Bitbucket, Jira, and Asana, through an event-driven architecture, automating cross-platform workflows for 40K+ monthly users
- Built an asynchronous webhook handling system with event deduplication to keep performance smooth during peak activity, reducing server load and unnecessary database writes
- Designed a normalized database with SQLAlchemy and OAuth 2.0 authentication, using indexing, token caching, and safe transactions to cut redundant queries, lower API usage costs, and ensure reliable data access
- Improved the workflow pipeline by integrating centralized error handling and detailed logging, which helped quickly identify failed events and prevent repetitive task retries

PROJECTS

GIT-RAG: RAG-based AI GitHub Repo Analysis Platform | [GitHub](#) | [Website](#)

- Enables fast analysis and understanding of large open-source GitHub repositories (up to 50 MB and 10,000+ files) through secure OAuth 2.0 login, robust API architecture and efficient large-scale code processing
- Resolved database storage limit issues from large repos by implementing a 3-stage RAG pipeline (Langchain) to offload large data as vectors to PineconeDB, preserving cloud PostgreSQL for smaller metadata
- Prevented database bloat and high costs by adding namespace cleanup to remove old repo data and limit each user to 1 active repo
- Engineered a parallel, chunk-based system that fixes slow and memory-heavy repo uploads, enabling efficient handling of large codebases and making chat responses faster and smoother

UNISPHERE: University Club and Community Hub | [GitHub](#) | [Website](#)

- Developed a unified platform with RESTful APIs and JWT authentication to replace manual, fragmented event and club management, enabling real-time admin control
- Structured unorganized user and event data with PostgreSQL schema using foreign keys, triggers, and indexes for fast, scalable access
- Optimized payments process through Stripe integration, automating registration, attendance, and finance tracking.
- Automated announcements via a smart Resend API email system with user-based filtering and real-time notifications through Ably for 1000+ users

ACHIEVEMENTS

- Solved over **1000 DSA questions** on LeetCode and Codeforces, achieving **1855 [Knight]** and **1293 [Pupil]** respectively
- **Won IEEE Open Source Week 9** by raising and merging **18 PRs**, solving issues ranging from simple UI fixes to complex ORM setup, database implementation, and CRUD operations
- Fixed UI/UX issues, bugs, and layout inconsistencies for GitHub projects like Bookingjini-Labs, UMatter, and Imagine_AI in GSSOC

CERTIFICATIONS

- AWS Educate Introduction to Cloud 101 (AWS) | [Link](#)
- Cloud Computing (NPTEL)
- The Bits and Bytes of Computer Networking (Google-Coursera)